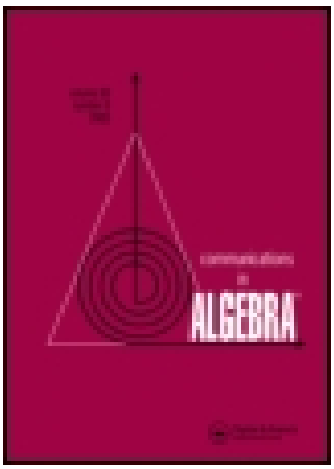
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Cohomology of algebras and triple systems

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COMMUNICATIONS IN ALGEBRA, 3(12), 1097-1120 (1975)

COHOMOLOGY OF ALGEBRAS AND TRIPLE SYSTEMS

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Introduction.

The classical cohomologies of unital associative algebras and of Lie algebras (cf.[2]) have both a double algebraic character: They are embedded in all of the machinery of derived functors, and they allow full extension theoretic interpretations (Yoneda-interpretation) of the higher cohomology groups - which seems natural since the coefficient category for cohomology is actually definable in terms of singular extension theory. If one wants to define a uniform cohomology theory for (linear) nonassociative algebras and triple systems which "generalizes" these two classical cohomologies one may proceed either via derived functors (cf.[1] and [8]) or via singular extension theory (cf.[3] or [4]). The purpose of this paper which adopts the first point of view is to discuss compatibility questions with the second one.

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We shall work over a ground field k. All tensor products are over k.

Representations and singular extensions of linear nonassociative algebras.

Let A be a linear nonassociative algebra of type I (where I means a given set of identities) over k. The pair (k,I) is always assumed to have the following property: $\ell \otimes B$ is of type I over ℓ for all I-algebras B over k and all unital commutative associative ring extensions $k \to \ell$. We shall identify the equivalent categories of I-bimodules for A, kernels of(short) singular I-extensions of A and unital left modules for "the" universal I-multiplication envelope U of A. For details we refer to [7]. (λ,ρ) will always mean the universal I-multiplication specialization from A to U. Given any I-bimodule M for A the canonical split extension of A by M and the singular I-extension defined by an I-factor set h for A in M will be denoted by $E(A,M)$ and $E_h(A,M)$ respectively.

For $n \geq 1$ let n-Ext(A,-) be the n-th Yoneda-functor (from I-bimodules for A to k-vector spaces). Thus 1-Ext(A,M) can be identified with Opext(A,M), the k-vector space of I-factor sets for A in M modulo trivial ones, this "parametrization" being induced by $h \to E_h(A,M)$. Furthermore consider a (short) generic I-extension

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$O \to K \to F \to A \to O$ of A. Then there are functorial isomorphisms

(*) $n\text{-}Ext(A,-) \simeq Ext_U^{n-1}(K,-)$, $n \geq 2$.

For details see [4].

Cohomology as the derived functor of derivations.

Let $\varphi : B \to A$ be a homomorphism of I-algebras over k, and let U be the universal I-multiplication envelope of A. $Der_\varphi(B,-)$, the functor of φ-derivations (from left U-modules to k-vector spaces), is representable, i.e. $Der_\varphi(B,-) \simeq Hom_U(D_\varphi(B),-)$ for some left U-module $D_\varphi(B)$ (cf. [1] or [8]). In particular ($\varphi = 1_A$) $Der(A,-) \simeq Hom_U(D(A),-)$, and we often will identify these functors.

$(H^n(A,-))_{n \geq o}$ will mean the sequence of right derived functors of Der(A,-), and will be identified with $(Ext_U^n(D(A),-))_{n \geq o}$. This will be the sequence of our n-th cohomology functors (from I-bimodules for A to k-vector spaces). Define a subfunctor Inder(A,-) of Der(A,-) by the following:

$d \in Inder(A,M)$ if and only if there is a locally finite family $(\partial_i)_{i \in \Lambda}$ in $Der(A,U)^\Lambda$ (for every $a \in A$ we have $\partial_i(a) = O$ for almost all $i \in \Lambda$) and a family $(m_i)_{i \in \Lambda}$ in M^Λ such that $d(a) = \Sigma \partial_i(a) m_i$ for all $a \in A$.

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Proposition 1: Der (A,-) is exact if and only if Der (A,-) = Inder (A,-).

Proof: $D(A) = U\otimes A/Q$, where Q is U-generated by $\{\lambda(a_1)\otimes a_2+\rho(a_2)\otimes a_1-1\otimes a_1a_2,\ a_1,a_2\in A\}$. Let $\partial:A \to D(A)$ be the universal derivation, i.e. $\partial(a)=1\otimes a \bmod Q$ for all $a\in A$. The exactness of Der(A,-) is clearly equivalent to the U-projectivity of D(A). But D(A) is U-projective if and only if $\partial\in$ Inder (A,D(A)):

For D(A) is U-projective if and only if there are families $(f_i)_{i\in\Lambda}$ in $\mathrm{Hom}_U(D(A),U)^\Lambda$ and $(c_i)_{i\in\Lambda}$ in $D(A)^\Lambda$ such that for every $c\in D(A)$ we have $f_i(c)=0$ for almost all $i\in\Lambda$ and $c=\Sigma f_i(c)c_i$. In particular $\partial(a)=\Sigma f_i\partial(a)c_i$ for all $a\in A$ and - with $\partial_i=f_i\partial$ - we have clearly $\partial\in$ Inder (A,D(A)). Conversely let ∂ be inner via appropriate $(\partial_i)_{i\in\Lambda},(c_i)_{i\in\Lambda}$ such that $\partial(a)=\Sigma\partial_i(a)c_i$ for all $a\in A$. Define $f_i\in\mathrm{Hom}_U(D(A),U)$ by $\partial_i=f_i\partial$ (universal factorization property of ∂), $i\in\Lambda$. Then we have for arbitrary $c\in D(A)$ ($\partial(A)$ generates D(A) as U-module!):

$$c = \Sigma u_k\partial(a_k)=\Sigma u_k\Sigma\partial_i(a_k)c_i=\Sigma\Sigma u_k f_i\partial(a_k)c_i =$$

$$\Sigma f_i(\Sigma u_k\partial(a_k))c_i = \Sigma f_i(c)c_i, \text{ and we are done.}$$

The universal factorization property of ∂ immediately yields the rest : If ∂ is inner, every

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derivation must be inner. (Inder (A,-) clearly means "inner derivation functor for A").

Remarks:

(i) Let A be finitely generated unital associative. Then Inder (A,-) is the classical inner derivation functor.

(ii) Let A be Lie or Jordan. Then the classical inner derivation functors are subfunctors of Inder (A,-), but do not coincide with Inder (A,-) in general.

The first cohomology group

A singular I-extension $E = O \to M \overset{\psi}{\to} B \overset{\varphi}{\to} A \to O$ of A gives rise to an exact sequence $M \overset{\partial\psi}{\to} D_{\varphi}(B) \overset{\partial\varphi}{\to} D(A) \to O$ of U-modules and U-homomorphisms (cf.[1]. Put $diff(E) = O \to M \overset{\partial\psi}{\to} D_{\varphi}(B) \overset{\partial\varphi}{\to} D(A) \to O$. We shall say that E allows a split embedding if and only if there is a singular I-extension $O \to N \to C \to A \to O$ of A which splits and such that McN, BcC and

$$\begin{array}{ccccccc} O & \to & M & \to & B & \to & A & \to & O \\ & & \cap & & \cap & & \| & & \\ O & \to & N & \to & C & \to & A & \to & O \end{array}$$

is commutative.

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Theorem: There is a k-injection $\Lambda_1 : H^1(A,M) \to 1\text{-Ext}(A,M)$ which is functorial in M and such that

$\mathrm{Im}\Lambda_1 = \{[E] : \mathrm{diff}(E) \text{ is exact}\} = \{[E] : E \text{ allows a split embedding}\}$

Proof:

1) Identifying $H^1(A,M)$ with $\mathrm{Ext}^1_U(D(A),M)$ and $1\text{-Ext}(A,M)$ with $\mathrm{Opext}(A,M)$, Λ_1 is obtained in the following way: Compute $\mathrm{Ext}^1_U(D(A),M)$ using the bar-resolution of $D(A) = U \otimes A/Q$,

i.e. $\mathrm{Ext}^1_U(D(A),M) = Z^1(D(A),M)/B^1(D(A),M)$, where

$Z^1(D(A),M) = \{g \in \mathrm{Hom}_k(U \otimes U \otimes A, M) : g(U \otimes Q) = 0$ and

$ug(v \otimes w \otimes a) - g(uv \otimes w \otimes a) + g(u \otimes vw \otimes a) = 0$

for all $u,v,w \in U, a \in A\}$

$B^1(D(A),M) = \{f \in \mathrm{Hom}_k(U \otimes U \otimes A, M) : f(U \otimes Q) = 0$ and there is a $f' \in \mathrm{Hom}_k(U \otimes A, M)$ satisfying $f'(Q) = 0$ and $f(u \otimes v \otimes a) = uf'(v \otimes a) - f'(uv \otimes a)$ for all $u,v \in U, a \in A\}$

Now a $g \in Z^1(D(A),M)$ defines an $h_g \in \mathrm{Hom}_k(A \otimes A, M)$ by $h_g(a_1 \otimes a_2) = g(\lambda(a_1) \otimes 1 \otimes a_2) + g(\rho(a_2) \otimes 1 \otimes a_1)$ for all $a_1, a_2 \in A$. h_g is an I-factor set for A in M, and Λ_1 is induced by $g \to h_g$ (for details see [8]).

2) In order to characterize the image of Λ_1 we first observe the following:

Let $E_h = E_h(A,M)$ be defined by the I-factor set h for A in M.

(i) $\mathrm{diff}(E_h)$ is exact if and only if there is a $\Theta\in \mathrm{Hom}_k(U\otimes A, M)$ such that the following holds:

$(L_{h,\Theta})$ $\quad uh(a_1\otimes a_2)=\Theta(u(\lambda(a_1)\otimes a_2+\rho(a_2)\otimes a_1-1\otimes a_1a_2))$ for all $a_1,a_2\in A, u\in U$.

(ii) $\mathrm{diff}(E_h)$ is exact and splits over U if and only if E_h is trivial.

Proof:

(i) Set $E_h: 0 \to M \xrightarrow{\psi} B \xrightarrow{\varphi} A \to 0$ for notational convenience. Consider $\partial\psi: M \xrightarrow{\tau} (U\otimes M)\oplus(U\otimes A) \xrightarrow{\mathrm{can}} D_\varphi(B)$, where $\tau(m)=1\otimes m$ for all $m\in M$.

$\partial\psi$ is injective if and only if there is a $\overline{\Theta}\in \mathrm{Hom}_k(U\otimes B, M)$ such that

$(\alpha)\ \overline{\Theta}(1\otimes m)=m$ for all $m\in M$, and $(\beta)\ \Theta(Q_\varphi)=0$, where $Q_\varphi=\mathrm{Ker}(U\otimes B \xrightarrow{\mathrm{can}} D_\varphi(B))$ is U-generated by $\{\lambda(\varphi b_1)\otimes b_2+\rho(\varphi b_2)\otimes b_1-1\otimes b_1b_2,\ b_1b_2\in B\}$.

(α) and (β) are easily seen to be equivalent to (α') and (β'), where

$(\alpha')\ \Theta(u\otimes m)=um$ for all $u\in U, m\in M$.

$(\beta')\ \overline{\Theta}(u(\lambda(a_1)\otimes a_2+\rho(a_2)\otimes a_1-1\otimes a_1a_2))=\overline{\Theta}(u\otimes h(a_1\otimes a_2))$ for all $a_1,a_2\in A, u\in U$.

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This implies:

$\partial\psi$ is injective if and only if there is a $\Theta\varepsilon Hom_k(U\otimes A,M)$ such that $(L_{h,\theta})$ holds.

For: $\overline{\Theta}$ satisfying (σ') and (β') yields $\Theta=\overline{\Theta}/U\otimes A$ satisfying $(L_{h,\theta})$. Conversely such a Θ can always be lifted according to (σ') and thus gives a $\overline{\Theta}$ satisfying (σ') and (β').

(ii) h is trivial if and only if there is a $\Theta\varepsilon Hom_U(U\otimes A,M)$ such that $(L_{h,\theta})$ holds. Furthermore diff (E_h) is exact and splits over U if and only if $\overline{\Theta}$ can be chosen out of $Hom_U(U\otimes B,M)$. Hence everything is clear via (i).

3) The characterization of the image of Λ_1 is now obtained by observing the following:

(i) Let h be an I-factor set for A in M such that $h=h_g$ for some $g\varepsilon Z^1(D(A),M)$. Define $\Theta\varepsilon Hom_k(U\otimes A,M)$ by $\Theta(u\otimes a)=g(u\otimes 1\otimes a)$. Then $(L_{h,\theta})$ holds.

(ii) Let h be an I-factor set for A in M and let $\Theta\varepsilon Hom_k(U\otimes A,M)$ be such that $(L_{h,\theta})$ holds. Define $g\varepsilon Hom_k(U\otimes U\otimes A,M)$ by $g(u\otimes v\otimes a)=\Theta(uv\otimes a)-u\Theta(v\otimes a)$. Then $g\varepsilon Z^1(D(A),M)$ and $h-h_g$ is trivial.

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(iii) In the situation of (ii) let $E_g(D(A),M)$ be the singular extension of $D(A)$ by M defined by g, and consider h as I-factor set for A in $E_g(D(A),M)$. Then $E_h(A,M)$ is embedded in the trivial extension $E_h(A,E_g(D(A),M))$, where h is trivialized by γ with $\gamma(a)=-(\partial(a)\oplus\theta(1\otimes a))$, $a\in A$.

(iv) Let $E_h(A,M)$ allow a split embedding. Then there is a $\theta\in Hom_k(U\otimes A,M)$ such that $(L_{h,\theta})$ holds.

Proof of (i) - (iv):

(i): $h(a_1\otimes a_2)=\theta(\lambda(a_1)\otimes a_2+\rho(a_2)\otimes a_1-1\otimes a_1a_2)$ by definition. Note that $\theta(1\otimes a_1a_2)=g(1\otimes 1\otimes a_1a_2)=0$ since $g\in Z^1(D(A),M)$. $\theta(u(\lambda(a_1)\otimes a_2+\rho(a_2)\otimes a_1-1\otimes a_1a_2))=u\theta(\lambda(a_1)\otimes a_2+\rho(a_2)\otimes a_1-1\otimes a_1a_2)$:

$\theta(u(\lambda(a_1)\otimes a_2))=u\theta(\lambda(a_1)\otimes a_2)+g(u\otimes\lambda(a_1)\otimes a_2)$

$\theta(u(\rho(a_2)\otimes a_1))=u\theta(\rho(a_2)\otimes a_1)+g(u\otimes\rho(a_2)\otimes a_1)$

$\theta(u\otimes a_1a_2)=u\theta(1\otimes a_1a_2)+g(u\otimes 1\otimes a_1a_2)$

and $g(U\otimes Q)=0$.

(ii) $g(uv\lambda(a_1)\otimes a_2)+g(uv\rho(a_2)\otimes a_1)-g(uv\otimes a_1a_2)=$

$\theta(uv\lambda(a_1)\otimes a_2)-u\theta(v\lambda(a_1)\otimes a_2)+\theta(uv\rho(a_2)\otimes a_1)-$

$u\theta(v\rho(a_2)\otimes a_1)-\theta(uv\otimes a_1a_2)+u\theta(v\otimes a_1a_2)=$

$(uv)h(a_1\otimes a_2)-u(vh(a_1\otimes a_2))=0$

whence $g(U\otimes Q)=0$.

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$ug(v\otimes w\otimes a)-g(uv\otimes w\otimes a)+g(u\otimes vw\otimes a)=0$

follows by write-out, and consequently $g\in Z^1(D(A),M)$. Finally the equivalence of h with h_g:

$h_g(a_1\otimes a_2)=g(\lambda(a_1)\otimes 1\otimes a_2)+g(\rho(a_2)\otimes 1\otimes a_1)=\theta(\lambda(a_1)\otimes a_2)$

$+\theta(\rho(a_2)\otimes a_1)-\theta(1\otimes a_1a_2)+\theta(1\otimes a_1a_2)-\lambda(a_1)\theta(1\otimes a_2)-$

$\rho(a_2)\theta(1\otimes a_1)=h(a_1\otimes a_2)+\theta(1\otimes a_1a_2)-a_1\theta(1\otimes a_2)-\theta(1\otimes a_1)a_2.$

(iii): Note that $E_g(D(A),M)=D(A)\oplus M$ as k-vector space, the U-module structure being given by $u(\partial(a)\oplus m)=u\partial(a)\oplus um+g(u\otimes 1\otimes a)$.

Now

$\gamma(a_1a_2)=-\partial(a_1a_2)\oplus-\theta(1\otimes a_1a_2)$

$-a_1\gamma(a_2)=a_1\partial(a_2)\oplus a_1\theta(1\otimes a_2)+g(\lambda(a_1)\otimes 1\otimes a_2)$

$-\gamma(a_1)a_2=\partial(a_1)a_2\oplus\theta(1\otimes a_1)a_2+g(\rho(a_2)\otimes 1\otimes a_1),$

thus γ trivializes h.

(iv) Let $E_h(A,M)$ be embedded in the trivial extension $0\to N\to C\to A\to 0$. Then there is some $\gamma\in Hom_U(U\otimes A,N)$ such that $h(a_1\otimes a_2)=\gamma(\lambda(a_1)\otimes a_2+\rho(a_2)\otimes a_1-1\otimes a_1a_2)$ for all $a_1,a_2\in A$. Let $p:N\to M$ be the k-linear projection. Set $\theta:=p\gamma$. Then $(L_{h,\theta})$ holds.

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This ends the proof of the theorem.

Corollary: Identifying $H^1(A,M)=Ext^1_U(D(A),M)=1\text{-}Ext(D(A),M)$ (Yoneda-interpretation in the module case) the inverse mapping of Λ_1 on $Im\Lambda_1$ is just induced by $E \to diff(E)$.

Proof: We are in the situation 3) (ii) of the proof above. We have to show: $E_g(D(A),M)$ is equivalent to $diff\,(E_h(A,M))$. Write $E_h(A,M)=0 \to M \to B \xrightarrow{\varphi} A \to 0$. The k-linear mapping $\bar{\sigma}:U\otimes A \to U\otimes B$ given by $\bar{\sigma}(u\otimes a)=u\otimes a-1\otimes\theta(u\otimes a)$ induces a k-section (for $\partial\varphi$) $\sigma:D(A) \to D_\varphi(B)$. Consider $D_\varphi(B)$ as the k-direct sum of $Ker\partial\varphi$ and $\sigma(D(A))$. The corresponding cocycle $g':U\otimes D(A) \to Ker\partial\varphi$ is defined by

$$g'(u\otimes(v\otimes a\bmod Q))=u\sigma(v\otimes a\bmod Q)-\sigma(uv\otimes a\bmod Q)=$$

$$uv\otimes a\bmod Q_\varphi-u\otimes\theta(v\otimes a)\bmod Q_\varphi-uv\otimes a\bmod Q_\varphi+1\otimes\theta(uv\otimes a)\bmod Q_\varphi$$

$$=1\otimes(\theta(uv\otimes a)-u\theta(v\otimes a))\bmod Q_\varphi$$

which means that $0 \to M \to D_\varphi(B) \xrightarrow{\partial\varphi} D(A) \to 0$ is equivalent to $E_g(D(A),M)$.

The higher cohomology groups

Consider a (short) generic I-extension $E_o=0 \to K \to F \xrightarrow{\pi} A \to 0$ of A with $diff(E_o)=0 \to K \to D_\pi(F) \xrightarrow{\partial\pi} D(A) \to 0$. This gives rise to an exact sequence $0 \to K_o \to K \xrightarrow{p} K_1 \to 0$, where $K_1=Ker\partial\pi$, and clearly also to a singular I-extension

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$E_1 = O \to K_1 \to F_1 \xrightarrow{\pi_1} A \to O$ (think in terms of factor sets: $E_o = E_h$, then $E_1 = E_{ph}$!), such that

$$\begin{array}{ccccccccc} O & \to & K & \to & F & \xrightarrow{\pi} & A & \to & O \\ & & \downarrow p & & \downarrow & & \| & & \\ O & \to & K_1 & \to & F_1 & \xrightarrow{\pi_1} & A & \to & O \end{array}$$

is commutative.

Proposition 2: For all $n \geq 2$ there are k-linear mappings $\Lambda_n : H^n(A,M) \to n\text{-Ext}(A,M)$, which are functorial in M.
$\mathrm{Im}\Lambda_n = \{[E] : [E]$ contains a representative of the form

$$O \to M \to \ldots \to X_1 \to F_1 \xrightarrow{\pi_1} A \to O\}.$$

Proof: $H^n(A,-) = \mathrm{Ext}^n_U(D(A),-) \simeq \mathrm{Ext}^{n-1}_U(K_1,-)$ because of the U-projectivity of $D_\pi(F)$ (cf. [4]).
$n\text{-Ext}(A,-) \simeq \mathrm{Ext}^{n-1}_U(K,-)$ by virtue of (*).
Hence Λ_n is essentially induced by p.
In order to get the description of the image of Λ_n we interpret Λ_n as mapping from $n\text{-Ext}(D(A),M)$ to $n\text{-Ext}(A,M)$. Λ_n is then induced by the following correspondence of n-fold singular extensions:
Represent $[E] \in n\text{-Ext}(D(A),M)$ by an element of the form $O \to M \to \ldots \to X_1 \to D_\pi(F) \xrightarrow{\partial\pi} D(A) \to O$ (take $D_\pi(F) \xrightarrow{\partial\pi} D(A) \to O$ as the beginning of a U-projective resolution for $D(A)$).
Consider the pull-back $X_1 \underset{K_1}{\times} K$. Then $\Lambda_n([E])$ is the equivalence class of

$$O \to M \to \ldots \to X_1 \underset{K_1}{\times} K \to F \xrightarrow{\pi} A \to O$$ and contains clearly

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the representative $O \to M \to \ldots \to X_1 \to F_1 \xrightarrow{\pi_1} A \to O$. Conversely, every $[E'] \in n\text{-}Ext(A,M)$ containing such a representative can be obtained in this way.

Remarks: (1) The following statements are equivalent:

(a) Λ_n is injective for all $n \geq 1$ and all M

(b) Λ_2 is injective for all M

(c) $O \to K_o \to K \xrightarrow{p} K_1 \to O$ splits over U.

(2) In the situation of (1) $diff(E_1)$ is exact, and one may identify $diff(E_1)$ with $O \to K_1 \to D_\pi(F) \xrightarrow{\partial\pi} D(A) \to O$. Then $E_1 = O \to K_1 \to F_1 \xrightarrow{\pi_1} A \to O$ is generic for all singular I-extensions E of A such that diff(E) is exact.

Singular extensions via derived functors

We shall say that an I-algebra A over k satisfies condition (L) if and only if diff(E) is exact for every singular I-extension E of A.

Proposition 3: The following statements are equivalent:

(1) A satifies (L).

(2) Every singular I-extension of A allows a split embedding.

(3) $diff(E_o)$ is exact for at least one generic I-extension E_o of A.

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(4) At least one generic I-extension of A allows a split embedding.

(5) Every singular I-extension of A by an injective I-bimodule R for A is trivial.

(6) For all $n \geq 1$ there are functorial isomorphisms $H^n(A,-) \simeq n\text{-}Ext(A,-)$.

Proof: The equivalence of (1) and (2) and of (3)and (4) is an immediate consequence of the theorem above. The equivalence of (1) and (3) follows from the observation: Given a generic I-extension $E_h*(A,K)$ of A, the genericity of $E_h*(A,K)$ can be expressed by the following genericity of h^*: For every I-factor set h for A in some I-bimodule M for A there is an $f \in Hom_U(K,M)$ such that $h - fh^*$ is trivial. Hence a $(L_{h^*,0^*})$- condition implies a $(L_{h,0})$- condition for any h (see part 2) of the proof of the theorem above), and we are done.

(2)implies (5): Embed the singular I-extension $0 \to R \to B \to A \to 0$ of A by R in a trivial extension $0 \to S \to C \to A \to 0$. Now R must be a U-direct factor in S, and thus the triviality "goes down".

(5) implies (2): Consider an arbitrary singular I-extension of A by an I-bimodule M for A. An embedding of M in an injective U-module R gives clearly rise to a split embedding of the singular extension in question.

(1) and (3) imply (6): $H^1(A,-) \simeq 1\text{-}Ext(A,-)$ by (1) and

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by the theorem above. By (3) diff(E_o) is exact for $E_o = O \to K \to F \xrightarrow{\pi} A \to O$ generic. Hence $K \simeq K_1 = \mathrm{Ker}\,\partial\pi$. Now (see proof of proposition 2) the functorial morphisms $\Lambda_n : H^n(A,-) \to n\text{-}\mathrm{Ext}(A,-)$, $n \geq 2$, are induced by $\mathrm{Ext}_U^{n-1}(K_1,-) \to \mathrm{Ext}_U^{n-1}(K,-)$ via $p: K \to K_1$ and are thus isomorphisms.

(6) implies clearly (5).

Corollary 1: Let $k \to \ell$ be a field extension. The A satisfies (L) if and only if $\ell \otimes A$ satisfies (L) (as an I-algebra over ℓ).

Proof: Let $O \to K \to F \to A \to O$ be the standard generic I-extension of A (cf.[4]). Then $O \to \ell \otimes K \to \ell \otimes F \to \ell \otimes A \to O$ is easily seen to be a generic I-extension of $\ell \otimes A$. A (generic) I-factor set h^* for A in K extends to a(generic) I-factor set $\bar{h}^*$ for $\ell \otimes A$ in $\ell \otimes K$. Since the universal I-multiplication envelope of $\ell \otimes A$ can be identified with $\ell \otimes U$(cf.[7]), a (L_{h^*,θ^*})-condition holds for h^* if and only if a $(L_{\bar{h}^*,\bar{\theta}^*})$-condition holds for $\bar{h}^*$.

Corollary 2: Let $\varphi: B \to A$ be a split epimorphism of I-algebras over k. Then A satifies (L) if B satifies (L).

Proof: Straightforward using $(L_{h,\theta})$- arguments.

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Corollary 3: The direct limit of I-algebras over k which all satisfy (L) is again an I-algebra over k satisfying (L).

Proof: Standard using $(L_{h,\theta})$-arguments. Note that universal I-multiplication envelopes commute with direct limits (cf.[7]).

Remarks:

(1) Neither homomorphic images nor subobjects of I-algebras satisfying (L) need again satisfy (L). See examples below.

(2) For an I-algebra A over k the following conditions are equivalent:

(i) Every singular I-extension of A is trivial.

(ii) At least one generic I-extension of A is trivial.

(iii) A satisfies (L) and Der(A,-)is exact.

Note that free I-algebras over k satisfy these conditions. Der(A,-) may be exact even if A does not satisfy (L); see example (c) below.

Some examples

It is not very difficult to see that one can obtain analogues results as above in the setting of unital cohomology and of cohomology of I-algebras with (anti-)

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automorphism of period 2 over a field k of characteristic ≠2 (cf.e.g.[4] for the definition of the relevant categories), since all considerations above carry over with the natural modifications. It is easy to show that an I-algebra with (anti-)automorphism of period 2 (A,σ) satisfies (L) if A satisfies (L) (considered merely as an I-algebra over k).

Now some examples:

(a) In the categories of unital associative algebras and of Lie algebras over k all objects satisfy (L). Let $\tilde{H}^n(A,-)$, $n\geq 0$, denote the usual cohomology functors for such an A (cf.[2]), then $H^n(A,-)\simeq\tilde{H}^{n+1}(A,-)$ for all $n\geq 1$ (cf.[1]).

(b) In the categories of unital associative algebras with antiautomorphism and of Lie algebras with automorphism of period 2 over a field k of characteristic ≠2 all ojects satisfy (L) by (a) and the foregoing remark. As to the Lie case, let $\tilde{H}^n((A,\sigma),-)$ $n\geq 0$, be the cohomology functors as defined by Harris (cf.[6]). Then $H^n((A,\sigma),-)\simeq\tilde{H}^{n+1}((A,\sigma),-)$ for all $n\geq 1$ by the same argument as in [1] (let U_σ be the universal Lie multiplication envelope "with automorphism of period 2" for (A,σ), and let $D(A,\sigma)$ be the module of differentials for (A,σ), then there

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is an exact sequence of U_σ-modules $0 \to D(A,\sigma) \to U_\sigma \to k \to 0$, and the long exact Ext-sequence provides the rest).

(c) In the category of unital commutative and associative algebras over k there exist objects which do not satisfy (L), e.g. inseparable field extensions A of appropriate k (cf.[4]). Note that for such an A Der (A,-) is clearly exact, and that for all $n \geq 2$ $H^n(A,-) \cong n\text{-Ext}(A,-) = 0$. The situation may become more pathological: see example (e) below.

(d) In the category of unital Jordan algebras over a field k of characteristic $\neq 2$ all Jordan matrix algebras of degree $n \geq 4$ satisfy (L)(cf.[5]). But there are also objects which do not satisfy (L): Consider $A = k[\varepsilon]$, the algebra of dual numbers over k, as unital Jordan algebra. The unital Jordan multiplication envelope U_1 of A can be identified with $k[\eta] = k[x]/_{(x^3)}$ (see[7]). U_1 is an injective U_1-module, since $U_1 \cong \text{Hom}_k(U_1,k)$ (as U_1-modules). The factor set h for A in U_1 given by $h(1\otimes 1) = h(1\otimes\varepsilon) = h(\varepsilon\otimes 1) = 0$, $h(\varepsilon\otimes\varepsilon) = 1$, is unital Jordan and non-trivial. Thus $H^1(A,U_1) = 0$ whereas $1\text{-Ext}(A,U_1) \neq 0$.

Consider now A as subalgebra of $H_4(A,\text{id})$ - the Jordan algebra of symmetric 4x4-matrices with coefficients in A. This is an example such that property (L) is

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not inherited by subobjects. Since free objects always satisfy (L) in the relevant category, it is now trivial that homomorphic images must not inherit (L).

(e) Finally we will show that the defect from satisfying (L) may or may not be reflected by the behaviour of the $\Lambda_n : H^n(A,-) \to n\text{-Ext}(A,-), n \geq 2$. Consider again $A=k[\varepsilon]$, the algebra of dual numbers over k, as well in the category of unital commutative and associative algebras (where it can be identified with U_1, its unital commutative and associative multiplication envelope) as in the category of unital Jordan algebras (see example (d) above). In both cases a unital generic extension of A is given by $0 \to K \to F \xrightarrow{\pi} A \to 0$, where $F=k[\xi]=k[x]/_{(x^4)}$, and $K=\text{Ker}\pi=(\xi^2)=k\xi^2 \oplus k\xi^3$. Clearly K is isomorphic to A, the regular U_1-module. Furthermore $D(A) \simeq k$ in both cases.

Now look first at the unital commutative and associative case: Here $K \simeq A=U_1$ is clearly U_1-projective, whence $\text{Ext}^n_{U_1}(K,-)=0$ for all $n \geq 1$ and thus $n\text{-Ext}(A,-)=0$ for all $n \geq 2$ by (*). But $H^n(A,k) \neq 0$ for all $n \geq 1$: $0 \to k \xrightarrow{f} U_1 \xrightarrow{g} k \to 0$ with $f1=\varepsilon$, $g1=1$, $g\varepsilon=0$, is an exact sequence of U_1-modules. Thus $\text{Ext}^n_{U_1}(k,-) \simeq \text{Ext}^{n+1}_{U_1}(k,-)$ for all $n \geq 1$, i.e.

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$H^n(A,-) \simeq H^{n+1}(A,-)$ for all $n \geq 1$. Clearly

$H^1(A,k) \neq 0$, since the exactness of

$$0 \to \mathrm{Hom}_{U_1}(k,k) \to \mathrm{Hom}_{U_1}(U_1,k) \to \mathrm{Hom}_{U_1}(k,k) \to 0$$

is impossible.

Now the unital Jordan case:

Consider the following two exact sequences of U_1-modules and U_1-homomorphisms:

$$0 \to A \xrightarrow{f} U_1 \xrightarrow{g} k \to 0$$

with $f'1=\eta$, $f'\varepsilon=\eta^2$, $g'1=1$, and

$$0 \to k \xrightarrow{f''} U_1 \xrightarrow{g''} A \to 0$$

with $f''1=\eta^2$, $g''1=1$, $g''\eta=\varepsilon$.

We then obtain functorial isomorphisms

$$\mathrm{Ext}^n_{U_1}(A,-) \simeq \mathrm{Ext}^{n+1}_{U_1}(D(A),-) \text{ and}$$

$$\mathrm{Ext}^n_{U_1}(D(A),-) \simeq \mathrm{Ext}^{n+1}_{U_1}(A,-) \text{ for all } n \geq 1,$$

and this implies by (*) $H^n(A,-) \simeq n\text{-}\mathrm{Ext}(A,-)$ for all $n \geq 2$ and $H^n(A,-) \simeq H^{n+2}(A,-)$ for all $n \geq 1$. Now $H^1(A,k) \neq 0$ by the same argument as above, and thus $H^n(A,k) \neq 0$ for all odd n.

Cohomology of triple systems

We shall briefly sketch the transcription of the results above to the case of triple systems over a field k.

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First the general representation theory of triple systems can be built up in complete analogy to the general representation theory of algebras as described in [7]: Starting with a set I of identities in the free k-triple system over a countable alphabet the category of I-triple systems over k is easily defined. As in the case of algebras the pair (k,I) will be assumed to be such as to guarantee type stability under scalar extensions.

Consider now a fixed I-triple system T over k. The coefficient category for the cohomology of T has again several equivalent descriptions: As category of I-trimodules for T (I-trirepresentations of T), as category of kernels of singular I-extensions of T or as category of unital left modules for "the" universal I-multiplication envelope U of T.

For $n\geq 1$ let n-Ext(T,-) be the n-th Yoneda-functor (from I-trimodules for T to k-vector spaces). Again 1-Ext(T,M) can be parametrized by the space of I-factor sets for T in M modulo trivial ones. Generic I-extensions $0 \to K \to F \to T \to 0$ of T exist as in the case of algebras (since free I-triple systems over k-spaces exist), and we can deduce again that $n\text{-Ext}(T,-)\cong \text{Ext}_U^{n-1}(K,-)$ for all $n\geq 2$.

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Let $\varphi: V \to T$ be a homomorphism of I-triple systems over k, and let $Der_{\varphi}(V,-)$ be the functor of φ-derivations (from I-trimodules for T to k-vector spaces), then $Der_{\varphi}(V,-) \simeq Hom_U(D_{\varphi}(V),-)$ for some left U-module $D_{\varphi}(V)$, in particular $(\varphi = 1_T)Der(T,-) \simeq Hom_U(D(T),-)$, and every singular I-extension $E = O \to M \to V \xrightarrow{\varphi} T \to O$ of T induces canonically $diff(E) = O \to M \to D_{\varphi}(V) \to D(T) \to O$, where everything is exact except possibly $O \to M \to D_{\varphi}(V)$.

Let $(H^n(T,-))_{n \geq o}$ be the sequence of right derived functors of the functor of derivations Der(T,-). This will be the sequence of our n-th cohomology functors for the cohomology of T.

Now all the results obtained for algebras can be translated to triple systems. In particular we can identify $H^1(T,-)$ with a subfunctor of 1-Ext(T,-) by the same characterization as in the case of algebras. Property(L) for triple systems is described in exactly the same manner as it is done for algebras in proposition 3.

Finally let us exhibit a category of triple systems with "good cohomology":

Proposition 4: Let k be a field of characteristic $\neq 2$. Let T be a Lie triple system over k.

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Then T satisfies (L).

Proof: Consider $(SL(T),\sigma)$, the standard Lie envelope (with automorphism of period 2 σ) of T. Let $E=E(T,M)$ be the canonical split extension of T by the Lie triple module M for T. SL(E) contains a σ-invariant ideal M_s such that $(SL(E),\sigma)$ can be identified with $E((SL(T),\sigma),(M_s,\sigma))$(cf.[6]). Note that $M=(M_s,\sigma)_-$, the k-space of skew elements in (M_s,σ). Consider now (M_s,σ) as module for $(UL(T),\sigma)$, the universal Lie envelope (with automorphism of period 2 σ) of T. Then there is the following result:

Let $0 \to M \to V \to T \to 0$ be a singular Lie triple extension of T, then there exists a singular Lie extension with automorphism of period 2

$0 \to (M_s,\sigma) \to (G,\sigma) \to (UL(T),\sigma) \to 0$ of $(UL(T),\sigma)$

such that $0 \to M \to V \to T \to 0$ is equivalent to $0 \to M \to (G,\sigma)_- \to T \to 0$ (cf.[6]). Now $(UL(T),\sigma)$ satisfies (L) by previous remarks, hence T satisfies (L) too (by proposition 3).

Note that there are analogues "going-up" constructions for singular extensions of Jordan algebras and Jordan triple systems to singular extensions of Lie algebras with automorphism of period 2, but the "going down" is by far less satisfactory.

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